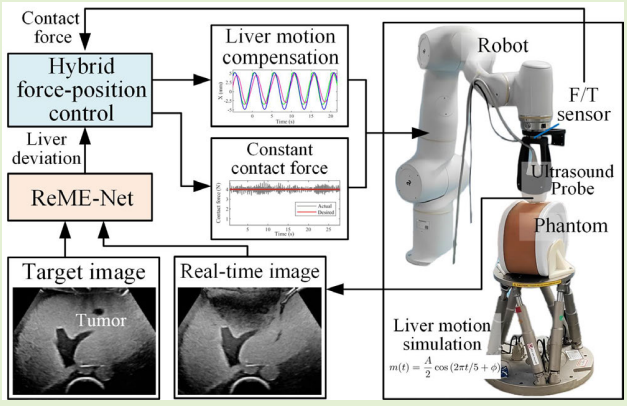
Robotic Respiratory Liver Motion Compensation for Stable Ultrasound Imaging of Small Lesions

Long Lei, Yingying Han, Jinhang Li, Ziwen Wang, Baoliang Zhao, Peng Zhang, and Ying Hu

***Abstract*—Real-time ultrasound-guided percutaneous liver biopsy plays a crucial role in diagnosing early liver cancer. However, patient respiration can cause liver drift, causing small target lesions to deviate from the ultrasound imaging plane and become invisible, complicating accurate punctures. To address this, we introduce an automatic robotic respiratory liver motion compensation method, which relies on real-time estimation of liver motion to maintain the stability between the ultrasound imaging plane and the lesion observation plane. We propose an end-to-end respiratory liver motion estimation network, termed ReME-Net. This network effectively utilizes the difference features between the current and target ultrasound frames, combined with the sequential channel-spatial attention mechanism, to accurately estimate the six-degree-of-freedom (6-DOF) respiratory liver motion in real time, without any tracking device. We employ a hybrid force–position control framework with an adaptive variable admittance controller to ensure stable probe-skin contact while compensating for liver motion, facilitating stable imaging of small lesions. On this basis, a robotic ultrasound imaging system is developed and experiments on a liver motion simulation platform demonstrate successful tracking with mean errors of 3.05 mm in translation, 2.52° in rotation, and 0.25 N in contact force, while periodic liver motion has a translation amplitude of 19.72 mm and a rotation amplitude of 17.32°. This method promises stable imaging of small liver lesions during breathing, which can improve early tumor puncture accuracy, helping medical professionals.**



***Index Terms*— Liver motion estimation, respiratory liver motion, robotic ultrasound imaging, stable contact control.**

I. Introduction

LIVER cancer is one of the most common malignant tumors worldwide, characterized by high incidence and high mortality rates. Early detection and treatment of liver cancer are crucial to improving patient survival rates and quality of life. Compared to CT and magnetic resonance imaging, ultrasound has the advantages of real-time imaging, nonradiation, cost-effectiveness, and portability, and ultrasound-guided percutaneous liver biopsy has been shown to be of significant value in the early diagnosis of liver cancer. During ultrasound-guided percutaneous puncture procedures, the surgeon presses the ultrasound probe on the patient's skin to image the target with one hand, and holds the needle in the other hand to puncture, while simultaneously looking at the ultrasound images. This process requires a high degree of hand-eye coordination and bilateral hand coordination. Furthermore, involuntary respiration by the patient can cause the liver to drift in the abdominal cavity, causing small tumor lesions to easily deviate from the ultrasound imaging plane and become invisible [1]. These challenges ultimately lead to a long learning curve for accurate ultrasound-guided puncture of small early stage tumors, and surgical results are highly dependent on the surgeon's experience. With the rapid development of robotics [2], robots can be employed to hold the ultrasound probe for automatic tracking and stable imaging of small tumors, thereby freeing surgeons from complicated dual-hand operation and allowing them to focus solely on

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puncture operation, and this solution has the potential to solve the problem of accurate puncture of small early-stage tumors in the liver under the influence of respiration.

Real-time and accurate estimation of the motion of the respiratory liver is a prerequisite for stable robotic ultrasound imaging of small lesions under the influence of respiration. The most common method is to model the respiratory motion between an easily measured surrogate signal and the internal target motion of interest based on their strong correlation, where the surrogate signal is measured in real time intraoperatively to estimate the target motion that is not readily available for direct measurement [3]. Popular surrogate signals include the volume of the tibial measured by spirometers or flowmeters [4], the physical movements of muscles near the lungs monitored by inertial measurement units [5] and accelerometers [6], and the displacement of the thoracic and abdominal cavities recorded by optical markers and other contactless methods [7]. Correlation models include polynomial models, B-spline models, and neural networks [8]. However, the time-varying nature of the patient's respiratory motion amplitude and frequency, coupled with intraoperative probe-tissue interactions, can cause the respiratory liver motion model established based on preoperative data to be inaccurate intraoperatively. In addition, modeling based on CT images requires multiple CT scans prior to surgery, which can cause additional radiation damage to the body.

To cope with time-varying respiratory motion, intraoperative lesion motion estimation based on real-time ultrasound image sequences has become a new research hotspot. The anatomical structural features and speckle information contained in ultrasound images provide the basis for respiratory-induced motion perception of tissue [9], [10]. In ultrasound imaging, the motion of a lesion can be decomposed into in-plane motion and out-of-plane motion. In-plane motion includes translation along the transverse and longitudinal directions as well as rotation around the normal direction, and out-of-plane motion includes translation along the normal direction as well as rotation around its transverse and longitudinal directions [11]. Anatomical structural characteristics are generally used to determine the in-plane movement of tissues. The ultrasound speckle of tissues is unique and stable, and the speckle correlation occurs for both in-plane and out-of-plane motion [9], so it can be used to track both in-plane and out-of-plane motion. Krupa et al. [12] first estimated the in-plane motion based on the classical template tracking technique to find the paired patches between images and then used an off-line calibrated speckle decorrelation model to compute the relative pose of six degrees of freedom (6-DOF) of speckle patches. However, traditional methods, especially speckle decorrelation methods, are complex and time-consuming and cannot meet the real-time requirements for clinical use.

Deep convolutional neural networks (CNNs) can automatically learn features that are decisive for the estimation of motion from high-dimensional data in an end-to-end manner and perform fast inference. For in-plane motion estimation, Evain et al. [13] implemented the estimation of myocardial motion in 2-D echocardiography based on the customized PWC-Net [14], which is a classical optical flow network framework. Liu et al. [15] proposed an attention-based hierarchical deep learning method and used transfer learning to detect the pose and mechanism state of a wireless capsule robot in ultrasound images. For out-of-plane motion estimation, Han et al. [16] proposed an attention-based network, named DAT-Net, this network takes ultrasound images with enhanced speckle noise features as input to calculate the out-of-plane distance between two consecutive ultrasound images under breathing. For the complete estimation of 6-DOF motion, some researchers used CNNs to directly estimate the relative pose of successive ultrasound frames, which are used to reconstruct the 3-D volume from free-hand 2-D ultrasound sequences without external tracking [17], [18], [19]. Prevost et al. [17] observed a strong correlation between CNNs and the speckle decorrelation process. For instance, local cross correlation is similar to convolution operations, and the selection of reliable speckle features can be achieved through activation layers. Therefore, they used a CNN to directly estimate the 6-DOF motion between adjacent ultrasound frames in an end-to-end manner. However, since the network structure was not specifically designed for the task, the accuracy of the estimation requires further improvement. Guo et al. [18] proposed a deep contextual learning network (DCL-Net) based on a self-attention mechanism. This network requires multiple consecutive ultrasound frames as input to efficiently utilize the hidden probe motion clues in the images. In our task, because of the presence of robotic respiratory liver motion compensation, the relative motion between the liver and the probe is a coupled result of the probe motion and the respiratory liver motion. Consequently, ultrasound image sequences provide limited effective contextual information about liver motion. Using ultrasound image sequences as input not only fails to improve accuracy, but also reduces inference speed. Hadramy et al. [19] proposed a method based on a recurrent neural network and Siamese architecture. However, this method requires complex preprocessing of ultrasound image sequences, including first obtaining sparse optical flow from the sequences and then generating Gaussian heatmaps of significant motion points. The Gaussian heatmaps, together with the ultrasound images, are input into the network to estimate the relative position of the frames. These preprocessing steps limit its application in real time. In addition, some researchers have integrated additional probe motion information into relative pose estimation networks for adjacent ultrasound images. This information is usually provided by external tracking devices attached to the probe [20] or by markers affixed to the skin [21]. Although such approaches can improve estimation performance, they also introduce additional costs and increase the complexity of both the model and the operation.

Unlike conventional ultrasound robots that scan according to a preplanned trajectory, stable ultrasound imaging of small lesions during free breathing requires that the robot be able to dynamically adjust the probe's posture based on both real-time lesion motion and probe-skin contact force. Among related studies, Seo et al. [22] calculated the target position by registering real-time biplane ultrasound images with a preoperative 3-D model and used a position-based

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visual servoing method to compensate for respiratory-induced lesion motion during high-intensity focused ultrasound (HIFU) therapy. Faoro et al. [23] used the optical flow method to estimate single-DOF respiratory motion within the ultrasound imaging plane and controlled a robotic arm for compensation. Krupa et al. [24] proposed a hybrid visual servoing method that consists of an out-of-plane 3-DOF motion servoing algorithm and a 2-D image-based in-plane 3-DOF visual servoing algorithm. However, they did not consider the contact force control between the probe and the skin. Nadeau and Krupa [11] applied an intensity gradient-based probe servoing method to track 5-DOF organ motion, while a proportional force control law was adopted to guarantee constant probe pressure against the skin. Some researchers have also used a position-based impedance control scheme to maintain an appropriate contact force between the probe and the patient's body during organ drift compensation using visual servoing [25] or remote control [26]. In addition, Jiang et al. [2], [27] employed an impedance controller based on built-in joint torque sensors to realize compliant contact control in Cartesian space. Wang et al. [28] proposed an admittance control-based hybrid force–velocity control framework to maintain stable contact force during robotic ultrasound scanning. These studies inform automated robotic small lesion stabilization ultrasound imaging control in free breathing. However, in these constant contact force controls based on impedance or admittance control models, constant control parameters are used, and the effect of tissue stiffness variation on the control performance is not considered.

To address the above problems, this article aims to accurately estimate the respiratory liver motion from real-time ultrasound images based on an end-to-end neural network, and on this basis, the robot holds the ultrasound probe for automatic respiratory-induced motion compensation, thereby achieving stable ultrasound imaging of small lesions, and eventually reducing the difficulty of ultrasound-guided percutaneous interventions. Our contributions are summarized as follows.

1) In order to accurately localize tumor lesions in real time, an end-to-end respiratory liver motion estimation network, termed ReME-Net, is proposed. This network leverages difference features between the current and target ultrasound images and employs sequential channel-spatial attention to estimate respiratory-induced 6-DOF liver motion.
2) To enable stable ultrasound imaging of small lesions under respiration, a robotic respiratory liver motion compensation control framework is designed, using online estimated respiratory-induced liver motion and probe-tissue contact force information to achieve adaptive compensation for liver movement and stable contact between the probe and the skin.
3) A robotic ultrasound imaging system is developed, and the experiments on a respiratory motion simulation platform demonstrate the accuracy and real-time performance of the respiratory liver motion compensation.

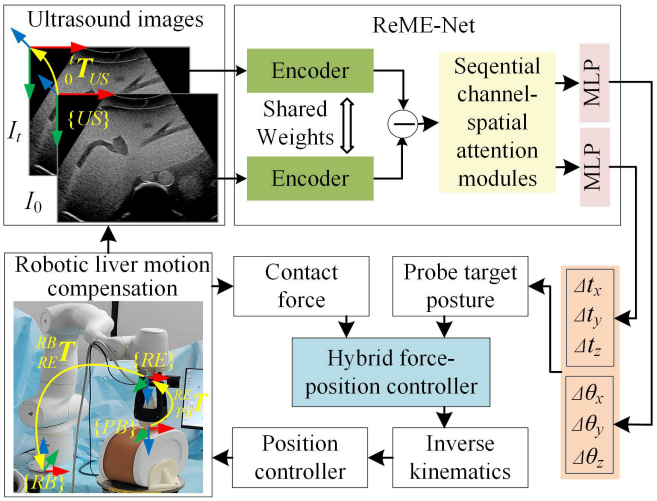


Fig. 1. Robotic autonomous 6-DOF liver motion compensation method based on online end-to-end respiratory liver motion estimation.

II. Materials and Methods

A. Robotic Liver Motion Compensation Framework

In order to ensure stable ultrasound imaging of small lesions under respiration, we propose a robotic autonomous 6-DOF liver motion compensation method using the online end-to-end respiratory liver motion estimation from real-time ultrasound images. As illustrated in Fig. 1, I_0 is the ultrasound image frame obtained with the probe in the initial ideal position, when the real-time ultrasound image I_t is captured, the two images, I_t and I_0, are fed into the end-to-end ReME-Net. This network outputs six transformation parameters $\{\Delta t_x, \Delta t_y, \Delta t_z, \Delta\theta_x, \Delta\theta_y, \text{ and } \Delta\theta_z\}$, which uniquely determine a homogeneous transformation matrix from the ideal imaging area to the actual imaging area of the liver, ${}^{t}_{0}\boldsymbol{T}_{US}$. Then, on the basis of the transformation ${}^{RB}_{US}\boldsymbol{T}_t$, which determines the actual imaging area of the liver in the robot base coordinate system $\{RB\}$ at time t, the ideal imaging area of the liver in the robot base coordinate system $\{RB\}$ at time t can be obtained by

$$ {}^{RB}_{US}\boldsymbol{T}'_t = {}^{RB}_{US}\boldsymbol{T}_t \cdot {}^{t}_{0}\boldsymbol{T}_{US} \tag{1} $$

$$ {}^{RB}_{US}\boldsymbol{T}_t = {}^{RB}_{RE}\boldsymbol{T}_t \cdot {}^{RE}_{US}\boldsymbol{T} \tag{2} $$

where ${}^{RB}_{RE}\boldsymbol{T}_t$ is the transformation matrix from the robot end flange coordinate system $\{RE\}$ to $\{RB\}$ at time t, which can be obtained by the robot forward kinematics. ${}^{RE}_{US}\boldsymbol{T}$ is the homogeneous transformation matrix from the ultrasound image coordinate system $\{US\}$ to $\{RE\}$. When the ultrasound imaging parameters (imaging depth and field of view) remain unchanged, this transformation is fixed. In this article, the transformation is calibrated using the N-wire model method from [1], with an rms error of 1.02 mm. To compensate for the deviation of the target imaging area caused by liver motion, the probe posture at time $t+1$ should be

$$ {}^{RB}_{PB}\boldsymbol{T}_{t+1} = {}^{RB}_{US}\boldsymbol{T}'_t \cdot \left({}^{RE}_{US}\boldsymbol{T}\right)^{-1} \cdot {}^{RE}_{PB}\boldsymbol{T} \tag{3} $$

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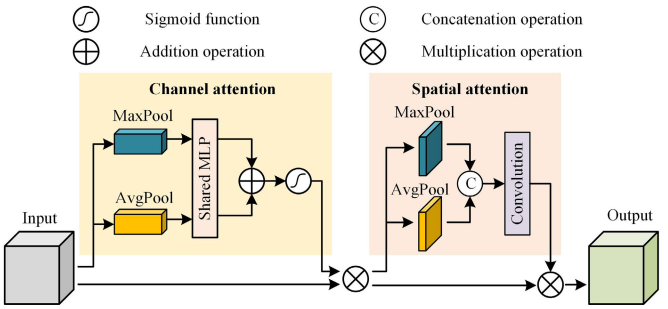


Fig. 2. Diagram of a sequential channel-spatial attention module.

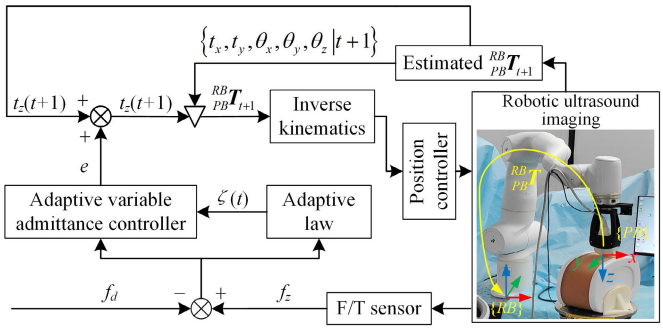


Fig. 3. Block diagram of the hybrid force–position control method.

where $\{PB\}$ is defined as the probe coordinate system with its origin at the intersection of the probe axis and the imaging surface, and the axes aligned with $\{RE\}$. ${}^{RE}_{PB}\boldsymbol{T}$ is also a calibrated transformation matrix from the ultrasound probe coordinate system $\{PB\}$ to $\{RE\}$.

Finally, using a hybrid force–position control method, the robot automatically adjusts the probe posture to compensate for liver motion while ensuring steady contact between the probe and tissue to maintain good imaging quality. In Sections II-B and II-C, we will introduce the proposed ReME-Net and position–force hybrid control algorithm in detail, respectively.

B. Respiratory Liver Motion Estimation Network

Inspired by DCL-Net [18], our ReME-Net adopts an encoder–decoder architecture. As shown in Fig. 1, two ultrasound images are inputted into the encoders with shared weights to obtain the corresponding feature maps. Then, the difference map between these two feature maps is computed using element-wise subtraction to highlight the cues correlated with the tissue motion. Finally, the difference map is fed into the decoder to infer six independent transformation parameters that describe the relative respiratory liver motion.

1) Encoder: In this work, to efficiently extract features that characterize tissue motion, such as anatomical structures and speckle features, from low-quality ultrasound images, a ResNeXt [29] module is used as the feature extractor. The ResNeXt module consists of multiple parallel branches that can accurately capture more diverse and detailed features without significantly increasing the parameters and maintaining computational efficiency. Specifically, a block of ResNeXt with two cardinalities is used. In addition, weight sharing between the two encoders is performed to decrease computational and storage requirements, thus speeding up the training and inference processes of the model. After the encoders, two 256-channel feature maps are obtained with the same width and height as the input images.

2) Decoder: After obtaining the difference feature map, we use the sequential channel-spatial attention module to further emphasize meaningful features and suppress unnecessary features. The sequential channel-spatial attention module is from the convolutional block attention module proposed by Woo et al. [30]. This module allows the channel attention branch to learn "what" to attend and the spatial attention branch to learn "where" to attend, respectively. As illustrated in Fig. 2, given the input feature map $\boldsymbol{F} \in \mathbb{R}^{C\times H\times W}$, the channel attention block infers a 1-D channel map $\boldsymbol{M}_c \in \mathbb{R}^{C\times 1\times 1}$, the spatial attention block infers a 2-D spatial attention map $\boldsymbol{M}_s \in \mathbb{R}^{1\times H\times W}$, and the final output of sequential channel-spatial attention can be summarized as

$$\begin{aligned}\boldsymbol{F}' &= \boldsymbol{M}_c(\boldsymbol{F}) \otimes \boldsymbol{F} \\ \boldsymbol{F}'' &= \boldsymbol{M}_s\left(\boldsymbol{F}'\right) \otimes \boldsymbol{F}'.\end{aligned} \tag{4}$$

In this work, three sequential channel-spatial attention modules are utilized one after another to understand the deep correlations between features more effectively to enhance the representation ability of the model. Then, the output features are flattened and two MLP blocks with one hidden layer and the ReLU activation function are used to regress the three translation parameters $\boldsymbol{\Delta t} = (\Delta t_x, \Delta t_y, \Delta t_z)$ and the three rotation parameters $\boldsymbol{\Delta r} = (\Delta\theta_x, \Delta\theta_y,$ and $\Delta\theta_z)$, respectively [31]. In addition, a Sage–Husa adaptive Kalman filter [32] is applied to adaptively correct the estimated parameters [33].

3) Loss: The whole ReME-Net is trained in a supervised manner. Due to the magnitude difference in the translation and rotation parameters, we divide the loss into translation loss and rotation loss components, and the mean squared error (mse) is used. The loss can be formulated as

$$\mathcal{L} = \alpha \cdot \text{mse}\left(\boldsymbol{\Delta t}, \boldsymbol{\Delta t}^*\right) + \beta \cdot \text{mse}\left(\boldsymbol{\Delta r}, \boldsymbol{\Delta r}^*\right) \tag{5}$$

where $\boldsymbol{\Delta t}^*$ and $\boldsymbol{\Delta r}^*$ are ground truth, and α and β are hyperparameters. In our implementation, we set $\alpha = 1$ and $\beta = 10$, respectively.

C. Hybrid Force–Position Control

While the robot moves the probe to the target posture ${}^{RB}_{PB}\boldsymbol{T}_{t+1}$, which corresponds to three rotational transformation parameters $\{\theta_x, \theta_y, \theta_z | t+1\}$ and three translational parameters $\{t_x, t_y, t_z | t + 1\}$, to compensate for the respiratory liver motion, a constant contact force must be maintained between the probe and the skin for stable imaging quality. For this reason, this article adopts the hybrid force–position control strategy, where the constant force is maintained along the probe axis and the estimated motion is compensated on the other five degrees of freedom. The hybrid force–position control framework is shown in Fig. 3. In particular, the surface geometry and the mechanical properties of the tissues, such as stiffness, vary at different locations of the abdomen, and to track the desired force quickly and accurately in the presence

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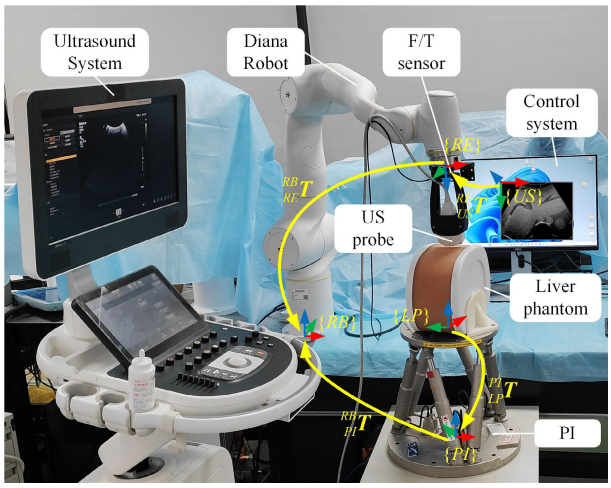


Fig. 4. Experimental setup for robotic respiratory liver motion compensation.

of uncertainty of stiffness, an adaptive variable admittance control method is used that uses online adjusted admittance parameters based on force tracking error [34]. The adaptive variable admittance control is illustrated as

$$M\ddot{e} + B(\dot{e} + \zeta) + Ke = f_z - f_d \tag{6}$$

where M, B, and K are the mass, damping, and spring constants, respectively. f_z represents the actual axial probe-tissue contact force, and f_d is the desired contact force. The value of f_d depends on factors such as the scanning area and the patient's obesity level. For individuals with more body fat, f_d can be increased to compress the fat layer, reducing the scattering and attenuation of the sound waves. For thinner individuals, where the fat layer is thinner and the probe can contact bones or hard tissues, excessive contact force can cause discomfort, pain, or artifacts in the image. Therefore, f_d should be reduced for thinner patients to avoid overcompression, with sufficient coupling gel used to minimize sound wave reflection and air interference. The variable $\zeta(t)$ changes over time and is adjusted online according to the force error (FE), as shown below

$$\zeta(t) = \zeta(t-1) + \eta\frac{f_d - f_z(t)}{B}, \quad \eta > 0 \tag{7}$$

where η is the update rate.

III. Experiments and Results

A. System Setup

To evaluate the methods proposed above, a robotic respiratory liver motion compensation system was developed for stable ultrasound imaging of small lesions, as shown in Fig. 4. The system consists of a six-axis robot (Diana 7 Med, Agile Robots, China), an ultrasound imaging system (Affiniti 30, Philips Inc., Holland) with a convex array probe (C6-4, Philips Inc., Holland), a frame grabber (OK_VGA41A, JoinHope Image Inc., China), a force/torque (F/T) sensor (mini40, ATI Industrial Automation, USA), and a computer host. Among them, the probe is connected to one end of the F/T sensor via a homemade clamp, and the other end of the F/T sensor is connected to the end flange of the robotic manipulator. The clamp is fabricated from Nylon 6 (PA6) using 3-D printing technology, resulting in a lightweight structure with relatively high stiffness. Finite element analysis indicates that its first natural frequency is 98.60 Hz, which is nearly five times higher than the control bandwidth. According to standard control theory, this ensures that the influence of the clamp dynamics on control stability and accuracy is negligible. The frame grabber is used to collect ultrasound images with a resolution of 800 × 600 pixels from the imaging system to the host control program in real time. The system control program is developed on the basis of the C++ language and open-source Qt library. In particular, the trained PyTorch-based respiratory liver motion estimation model is deployed into the software based on LibTorch. The whole control system can achieve a stable refresh frequency of 20 Hz. To evaluate the methods proposed above, a robotic respiratory liver motion compensation system was developed for stable ultrasound imaging of small lesions, as shown in Fig. 4. The system consists of a six-axis robot (Diana 7 Med, Agile Robots, China), an ultrasound imaging system (Affiniti 30, Philips Inc., Holland) with a convex array probe (C6-4, Philips Inc., Holland), a frame grabber (OK_VGA41A, JoinHope Image Inc., China), a F/T sensor (mini40, ATI Industrial Automation, USA), and a computer host. Among them, the probe is connected to one end of the F/T sensor via a homemade clamp, and the other end of the F/T sensor is connected to the end flange of the robotic manipulator. The frame grabber is used to collect ultrasound images with a resolution of 800 × 600 pixels from the imaging system to the host control program in real time. The system control program is developed on the basis of the C++ language and open-source Qt library. In particular, the trained PyTorch-based respiratory liver motion estimation model is deployed into the software based on LibTorch. The whole control system can achieve a stable refresh frequency of 20 Hz.

In addition, an abdominal phantom (CIRS 057A, Sun Nuclear, USA) is mounted on the moving platform of a hexapod microrobot (H-811, Physik Instrumente, USA), and each of the 6-DOF of its moving stage is given a periodic motion [1, eq. (8)] with different amplitudes and initial phases to simulate liver motion under breathing. Specifically, the three rotational degrees of freedom are described using Euler angles in the XYZ order with respect to fixed axes. These angles define the transformation of a unit vector initially aligned with the Z-axis, providing a precise representation of the phantom's orientation in space

$$m(t) = \frac{A}{2}\cos(2\pi t/5 + \phi) \tag{8}$$

where A represents the motion amplitude and ϕ represents the initial phase.

B. Data Collection

In order to train the end-to-end respiratory motion estimation model, pairs of ultrasound images with true relative motion in the ultrasound image coordinate system should be collected. The coordinate transformations involved in the

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experimental setup are illustrated in Fig. 4. Among them, $\{PI\}$ represents the base coordinate system of the hexapod microrobot. The liver phantom coordinate system $\{LP\}$ is set to coincide with the microrobot moving platform coordinate system. The inferior-to-superior direction of the phantom is aligned with the X-axis, and the right-to-left direction is aligned with the Y-axis. This setup allows the transformation ${}^{PI}_{LP}\boldsymbol{T}$ to be obtained directly from the robot forward kinematics. The transformation ${}^{RB}_{PI}\boldsymbol{T}$ is calibrated before mounting the probe and the phantom using the dual-arm calibration method described in [1]. During data collection, the relative position between the base of the hexapod microrobot and the six-axis robot base must remain fixed to ensure that the calibrated transformation ${}^{RB}_{PI}\boldsymbol{T}$ remains valid. The calibration process is roughly as follows: two specific steel styli are mounted first on the end of the six-axis robot and the hexapod microrobot, respectively. Then, using the four-point method, the tip of each stylus is defined as the tool center point (TCP) of the six-axis robot and the hexapod microrobot. Finally, by bringing the tips of the two styli into contact with four asymmetrical points in space, the positions of these contact points are recorded in the coordinate systems $\{RB\}$ and $\{PI\}$, respectively. A system of equations is formulated based on these pairwise positional data, and solving this system provides the desired transformation. In this article, the rms error of the calibration is 0.39 mm. Based on these aforementioned transformations, the liver phantom motion can be mapped to the ultrasound image coordinate system $\{US\}$ as

$$ {}^{US}_{LP}\boldsymbol{T} = \left({}^{RE}_{US}\boldsymbol{T}\right)^{-1} \cdot \left({}^{RB}_{RE}\boldsymbol{T}\right)^{-1} \cdot {}^{RB}_{PI}\boldsymbol{T} \cdot {}^{PI}_{LP}\boldsymbol{T}. \tag{9} $$

For data collection, the liver phantom moves periodically according to different predefined trajectories, during which the robot holds the ultrasound probe and enables adaptive variable admittance control to ensure stable contact between the probe and the liver phantom in the Z-direction, while the other degrees of freedom remain stationary, and ultrasound images and the corresponding transformations ${}^{US}_{LP}\boldsymbol{T}$ are saved simultaneously. Then, for adjacent ultrasound frames I_i and I_{i+1}, their true relative transformation ${}^{i}_{i+1}\boldsymbol{T}_{US}$ in the ultrasound image coordinate system can be calculated by

$$ {}^{i}_{i+1}\boldsymbol{T}_{US} = {}^{US}_{LP}\boldsymbol{T}_i \cdot \left({}^{US}_{LP}\boldsymbol{T}_{i+1}\right)^{-1}. \tag{10} $$

In this way, pairs of adjacent ultrasound images with real relative motion can be collected. In this study, data from 37 motion trajectories were collected, with each trajectory consisting of eight cycles. For these trajectories, the translational amplitude in the X-direction (A_x) was set to integer values between 9 and 15 mm, the translational amplitude in the Y-direction (A_y) was set to integer values between 6 and 10 mm, and the translational amplitude in the Z-direction (A_z) was set to integer values between 4 and 8 mm. For each trajectory, the translation amplitude in the X-direction was greater than in the Y-direction, and the amplitude in the Y-direction was greater than in the Z-direction. The amplitudes of three rotational degrees of freedom (A_{rx}, A_{ry}, A_{rz}) were set at 10° for all trajectories. These motion amplitudes follow the characteristics of real human liver motion as reported by Bertholet et al. [35]. The initial phase of each degree of freedom for all motion trajectories was set to a random value between 0 and 1.25 radians (a quarter of a cycle). Finally, 100 image pairs were extracted from each trajectory and the collected data were divided according to motion trajectories: data from 27 trajectories were used for training, five for validation, and the remaining five for testing. The five trajectories in the test set are shown in Fig. 5.

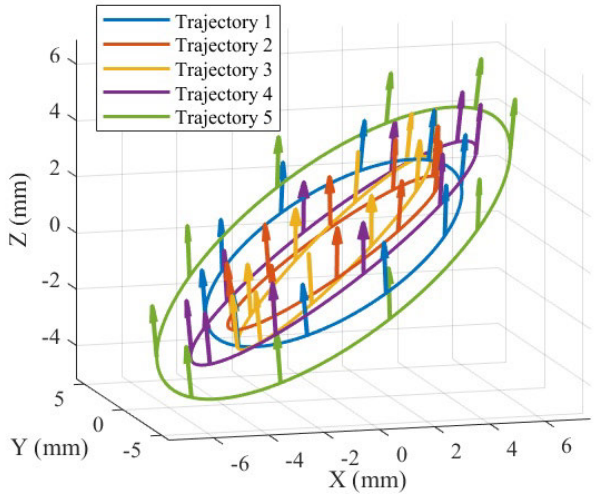


Fig. 5. Five motion trajectories of the phantom in the test set, with arrows indicating the directions at the sampled points.

TABLE I
COMPARATIVE RESULTS OF DIFFERENT MODELS IN TERMS OF FOUR QUANTITATIVE METRICS: MEAN±STD

Models	TE (mm)	RE (○)	TDD (mm)	Run time (FPS)
DCL-Net	2.56±1.35	1.93±1.11	3.29±1.62	4
Our ReME-Net	1.73±1.03	1.20±0.60	1.88±1.10	32

C. Results and Analysis

1) ReME-Net: To show the performance of our model, we compared it with DCL-Net [18], which was modified to estimate the transformation parameters between the current image and the initial ideal image rather than the average parameters over multiple images. For evaluation, we adopt the translation error (TE) to represent the $L2$ distance between the true and estimated translation vectors (Δt_x, Δt_y, Δt_z), and the rotation error (RE) to represent the $L2$ distance between the true and estimated rotation vectors ($\Delta\theta_x$, $\Delta\theta_y$, $\Delta\theta_z$). The target drift distance (TDD) denotes the Euclidean distance between the center points of the current frame and transformed ideal frame by the estimated relative motion [18]. In addition, the run times are also reported to reveal the motion estimation efficiency of the motion estimation networks. As illustrated in Table I, our model significantly outperforms DCL-Net for registration accuracy, for example, about 42.86% improvement for the TDD, which can be attributed to the interaction of difference features from input frames by the sequential channel-spatial attention mechanism. For motion estimation efficiency, our ReME-Net model is significantly faster than the DCL-Net (requiring multiple ultrasound frames

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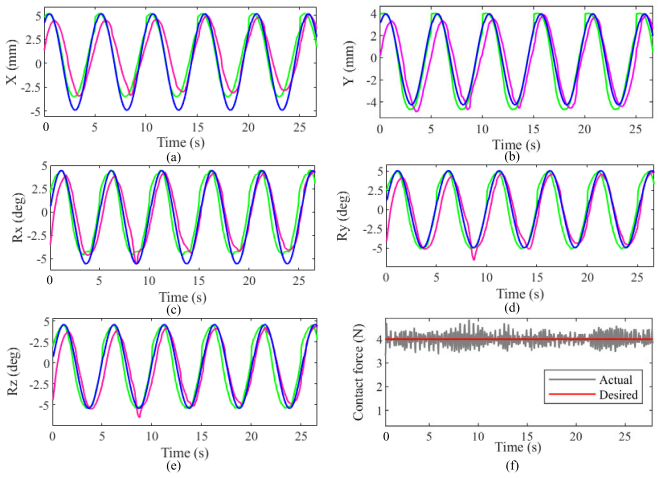


Fig. 6. Visualization results of robotic respiratory liver motion compensation for (a) translation along the *X*-axis, (b) translation along the *Y*-axis, (c) rotation around the *X*-axis, (d) rotation around the *Y*-axis, (e) rotation around the *Z*-axis, and (f) contact force in the *Z*-axis direction. In (a)–(e), pink curves illustrate robotic respiratory liver motion compensation results, the green curves show the estimation results using our ReME-Net, while the blue curves represent the given liver motion. For ease of comparison, all data in (a)–(e) have been mean-centered.

TABLE II
EXPERIMENTAL RESULTS OF ROBOTIC RESPIRATORY LIVER MOTION TRACKING

Trajectory	1	2	3	4	5
Translation amplitude (mm)	13.75	11.87	13.04	16.73	19.72
Mean TTE (mm)	2.49	1.73	1.86	2.60	3.05
STD TTE (mm)	1.66	0.98	1.20	1.18	1.77
Rotation amplitude (°)	17.32	17.32	17.32	17.32	17.32
Mean RTE (°)	2.28	2.41	2.19	2.32	2.52
STD RTE (°)	1.07	1.08	1.06	1.03	1.10
Desired force (N)	4.00	4.00	4.00	4.00	4.00
Mean FE (N)	0.13	0.14	0.14	0.13	0.25

as input and using case-wise correlation loss), further confirming the superiority of our ReME-Net model in real-time respiratory liver motion estimation.

2) Robotic Respiratory Liver Motion Compensation Control: In the robotic autonomous compensatory control experiments, the liver followed five trajectories corresponding to the test set in the Section III-B, and the experimental results of the first trajectory are presented in Fig. 6. As observed in the figure, the respiratory liver motion estimated by our ReME-Net model is able to match the trajectories of the real liver motion well. For robotic liver motion compensation, although there is a slight time delay mainly caused by the motion compensation strategy and the actuator response time, the probe tracks the liver motion well. In addition, under the dynamic liver motion conditions, the fluctuation range of the force control remains below 1 N.

Table II shows quantitative results of the robotic respiratory liver motion compensation over eight respiratory motion cycles for the five liver motion trajectories. In the table, the translation amplitude is the norm of the vector (A_x, A_y, A_z), where A_x, A_y, and A_z are amplitudes in X-, Y-, and Z-direction translation in (8), respectively, and the translation tracking error (TTE) represents the $L2$ distance between the ideal and actual translation vectors (t_x, t_y). The rotation amplitude is the norm of the vector (A_{rx}, A_{ry}, A_{rz}), where A_{rx}, A_{ry}, and A_{rz} are amplitudes around X-axis, Y-axis, and Z-axis rotation in (8), respectively, and the rotation tracking error (RTE) represents the $L2$ distance between the ideal and actual rotation vectors $(\theta_x, \theta_y, \theta_z)$. In addition, desired contact force and FE in Z-direction are also listed. It can be seen that in the five experiments, the mean force control error ranged from 0.13 to 0.25 N, while the mean TTE ranged from 1.73 to 3.05 mm and the mean RTE ranged from 2.19° to 2.52°. In the last experiment, both the force control and position control errors were higher than others, this is because the liver has the largest movement amplitude in this experiment, resulting in the fastest movement, under the same robot control frequency, the position and force tracking effects inevitably become worse.

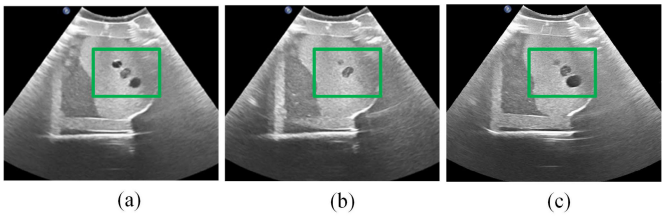


Fig. 7. Ultrasound imaging results of target anatomical structure marked with a green wireframe (a) in desired condition, (b) with only constant force control without motion compensation, and (c) with robotic liver motion compensation.

Fig. 7 shows the tracking effect of the target anatomical structures with and without robotic liver motion compensation for the first trajectory when the liver deviates the most from the initial position. It is obvious that the target anatomical structures are better maintained under the ultrasound image field of view in the presence of robotic liver motion compensation.

IV. CONCLUSION

This article presents a robotic autonomous liver motion compensation method based on online estimation of respiratory liver motion, achieving stable ultrasound imaging of the target small lesions during free breathing. Real-time and accurate estimation of respiratory liver motion is achieved using a lightweight end-to-end neural network model, termed ReME-Net. This model takes the current and ideal ultrasound images as input and leverages a sequential channel spatial attention mechanism to extract meaningful features and to estimate the relative motion between the two images. Comparative experiments demonstrate that our ReME-Net outperforms the current state-of-the-art model in both precision and efficiency. Liver motion tracking is performed using a hybrid force–position control framework, where an adaptive variable admittance control is applied to ensure constant contact force between the probe and the tissue in the presence of uncertainty of tissue stiffness. Based on the proposed methods, a robotic ultrasound imaging system is developed, and experiments on a respiratory liver motion simulation platform show that stable ultrasound imaging of the target small lesion under

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the influence of breathing can be effectively achieved. For example, for periodic liver motion with a translation amplitude of 19.72 mm and a rotation amplitude of 17.32°, the mean translation and RTEs are 3.05 mm and 2.52°, respectively, while maintaining a mean contact force control error of 0.25 N.

A major limitation of this work is that, for convenience in data collection, we performed only experimental validation on an abdomen phantom. In the phantom, the anatomical structures are simplified, the organ tissues are homogeneous, and there is a significant acoustic impedance contrast between different tissues. However, the actual human physiological environment is far more complex, and real patient imaging often involves significant challenges due to ultrasound artifacts. Typically, artifacts such as acoustic shadowing or enhancement can obscure critical anatomical structures, leading to incomplete or inaccurate motion estimation and compensation. Mirror imaging artifacts can introduce false structures or duplicate existing ones, potentially confusing motion estimation algorithms [36]. Therefore, the proposed method in this article requires further validation on human subjects and improvements to address the complexities of the human environment and ultrasound imaging artifacts. In addition, in the proposed robotic respiratory liver motion compensation method, the network estimates the liver's already occurred movement, and combined with the control signal processing delay and the robot's physical execution delay, it inevitably leads to a delay in the robot's autonomous respiratory motion compensation. Therefore, future work will incorporate a respiratory motion prediction function to predict the liver's motion state at the next moment in advance. This will enable the robot to compensate preemptively, offsetting the execution delay and achieving higher accuracy in respiratory motion tracking performance.

REFERENCES

[1] L. Lei et al., "Robotic needle insertion with 2D ultrasound–3D CT fusion guidance," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 4, pp. 6152–6164, Apr. 2023.

[2] Z. Jiang, S. E. Salcudean, and N. Navab, "Robotic ultrasound imaging: State-of-the-art and future perspectives," *Med. Image Anal.*, vol. 89, Oct. 2023, Art. no. 102878.

[3] L. Lei et al., "Diffeomorphic respiratory motion estimation of thoracoabdominal organs for image-guided interventions," *Med. Phys.*, vol. 48, no. 8, pp. 4160–4176, Aug. 2021.

[4] C. Massaroni et al., "Respiratory monitoring during physical activities with a multi-sensor smart garment and related algorithms," *IEEE Sensors J.*, vol. 20, no. 4, pp. 2173–2180, Feb. 2020.

[5] T. Elfaramawy, C. L. Fall, S. Arab, M. Morissette, F. Lellouche, and B. Gosselin, "A wireless respiratory monitoring system using a wearable patch sensor network," *IEEE Sensors J.*, vol. 19, no. 2, pp. 650–657, Jan. 2019.

[6] S. Fahmi, F. F. J. Simonis, and M. Abayazid, "Respiratory motion estimation of the liver with abdominal motion as a surrogate," *Int. J. Med. Robot. Comput. Assist. Surg.*, vol. 14, no. 6, p. e1940, Dec. 2018.

[7] C. Massaroni, A. Nicolo, M. Sacchetti, and E. Schena, "Contactless methods for measuring respiratory rate: A review," *IEEE Sensors J.*, vol. 21, no. 11, pp. 12821–12839, Jun. 2021.

[8] V. Spieker et al., "Deep learning for retrospective motion correction in MRI: A comprehensive review," *IEEE Trans. Med. Imag.*, vol. 43, no. 2, pp. 846–859, Feb. 2024.

[9] C. B. Burckhardt, "Speckle in ultrasound B-mode scans," *IEEE Trans. Sonics Ultrason.*, vol. SU-25, no. 1, pp. 1–6, Jan. 1978.

[10] P. Abolmaesumi, S. E. Salcudean, W.-H. Zhu, M. R. Sirouspour, and S. P. DiMaio, "Image-guided control of a robot for medical ultrasound," *IEEE Trans. Robot. Autom.*, vol. 18, no. 1, pp. 11–23, Jan. 2002.

[11] C. Nadeau and A. Krupa, "Intensity-based ultrasound visual servoing: Modeling and validation with 2-D and 3-D probes," *IEEE Trans. Robot.*, vol. 29, no. 4, pp. 1003–1015, Aug. 2013.

[12] A. Krupa, G. Fichtinger, and G. D. Hager, "Real-time tissue tracking with B-mode ultrasound using speckle and visual servoing," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, Oct. 2007, pp. 1–8.

[13] E. Evain et al., "Motion estimation by deep learning in 2D echocardiography: Synthetic dataset and validation," *IEEE Trans. Med. Imag.*, vol. 41, no. 8, pp. 1911–1924, Aug. 2022.

[14] D. Sun, X. Yang, M.-Y. Liu, and J. Kautz, "PWC-net: CNNs for optical flow using pyramid, warping, and cost volume," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8934–8943.

[15] X. Liu et al., "Capsule robot pose and mechanism state detection in ultrasound using attention-based hierarchical deep learning," *Sci. Rep.*, vol. 12, no. 1, p. 21130, Dec. 2022.

[16] Y. Han et al., "A learning-based respiratory motion estimation with consecutive ultrasound images," in *Proc. IEEE Int. Conf. Imag. Syst. Techn. (IST)*, Oct. 2023, pp. 1–6.

[17] R. Prevost et al., "3D freehand ultrasound without external tracking using deep learning," *Med. Image Anal.*, vol. 48, pp. 187–202, Aug. 2018.

[18] H. Guo, S. Xu, B. J. Wood, and P. Yan, "Sensorless freehand 3D ultrasound reconstruction via deep contextual learning," in *Proc. Med. Image Comput. Comput. Assist. Intervent.*, 2020, pp. 463–472.

[19] S. E. Hadramy, J. M. Verde, K.-P. Beaudet, N. Padoy, and S. Cotin, "Trackerless volume reconstruction from intraoperative ultrasound images," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2023, pp. 303–312.

[20] M. Luo et al., "Multi-IMU with online self-consistency for freehand 3D ultrasound reconstruction," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2023, pp. 342–351.

[21] L. Dai, K. Zhao, Z. Li, J. Zhu, and L. Liang, "Advancing sensorless freehand 3D ultrasound reconstruction with a novel coupling pad," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2024, pp. 559–569.

[22] J. Seo, N. Koizumi, M. Mitsuishi, and N. Sugita, "Ultrasound image based visual servoing for moving target ablation by high intensity focused ultrasound," *Int. J. Med. Robot. Comput. Assist. Surg.*, vol. 13, no. 4, p. 1793, Dec. 2017.

[23] G. Faoro, N. Pasini, A. Mariani, L. Morchi, S. Tognarelli, and A. Menciassi, "Robot-assisted motion compensation based on optical flow in ultrasound images," in *Proc. IEEE Int. Ultrason. Symp. (IUS)*, Sep. 2023, pp. 1–4.

[24] A. Krupa, G. Fichtinger, and G. D. Hager, "Real-time motion stabilization with B-mode ultrasound using image speckle information and visual servoing," *Int. J. Robot. Res.*, vol. 28, no. 10, pp. 1334–1354, Oct. 2009.

[25] I. Paranawithana et al., "Ultrasound-guided involuntary motion compensation of kidney stones in percutaneous nephrolithotomy surgery," in *Proc. IEEE 14th Int. Conf. Autom. Sci. Eng. (CASE)*, Aug. 2018, pp. 1123–1129.

[26] S. Ipsen, D. Wulff, I. Kuhlemann, A. Schweikard, and F. Ernst, "Towards automated ultrasound imaging—Robotic image acquisition in liver and prostate for long-term motion monitoring," *Phys. Med. Biol.*, vol. 66, no. 9, May 2021, Art. no. 094002.

[27] Z. Jiang et al., "Autonomous robotic screening of tubular structures based only on real-time ultrasound imaging feedback," *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7064–7075, Jul. 2022.

[28] Z. Wang et al., "Full-coverage path planning and stable interaction control for automated robotic breast ultrasound scanning," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7051–7061, Jul. 2023.

[29] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.

[30] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[31] L. Lei et al., "Epicardium prompt-guided real-time cardiac ultrasound frame-to-volume registration," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2024, pp. 618–628.

[32] A. P. Sage and G. W. Husa, "Adaptive filtering with unknown prior statistics," in *Proc. Joint Autom. Control Conf.*, 1969, no. 7, pp. 760–769.

[33] W. Yan, Q. Ding, J. Chen, Y. Liu, and S. S. Cheng, "Needle tip tracking in 2D ultrasound based on improved compressive tracking and adaptive Kalman filter," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3224–3231, Apr. 2021.

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[34] J. Duan, Y. Gan, M. Chen, and X. Dai, "Adaptive variable impedance control for dynamic contact force tracking in uncertain environment," *Robot. Auto. Syst.*, vol. 102, pp. 54–65, Apr. 2018.

[35] J. Bertholet, E. S. Worm, W. Fledelius, M. Høyer, and P. R. Poulsen, "Time-resolved intrafraction target translations and rotations during stereotactic liver radiation therapy: Implications for marker-based localization accuracy," *Int. J. Radiat. Oncol. Biol. Phys.*, vol. 95, no. 2, pp. 802–809, Jun. 2016.

[36] S. Abbas and P. Peng, "Basic principles and physics of ultrasound," in *Ultrasound for Interventional Pain Management: An Illustrated Procedural Guide*, 2020, pp. 1–31.



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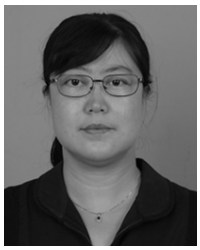
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